

satiation

Human Satiation & Completeness in AI Economics — Research Study

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Research study · draft · OpenIE research.openie.dev

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Honesty. Research study / draft paper — not a final journal PDF. No fabricated citations. Claims keep honesty tags ([UNVERIFIED] / [VENDOR] / [SURROGATE] / [SOFT] as present in the study).
board_synth_claimed=false — software reference stack; surrogate joules ≠ board watts.

Capability scales. Appetite does not. Design to done.

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Role: Comprehensive research layer backing Medium #2 (MEDIUM_SATIATION) and a future whitepaper; decision rules already in SATIATION_BLUEPRINT (research archive).

Honesty: No fabricated citations. Soft historical analogies labeled [SOFT]. Unverified or vendor claims marked [UNVERIFIED] / [VENDOR] / [SURROGATE]. Stack remains software reference: board_synth_claimed=false.

Companion artifacts: PRODUCT_LANDSCAPE, NOTATIONAL_INTELLIGENCE_RESEARCH, PRODUCT, SYSTEM_ONE_TO_WCA (research archive).

0. Executive synthesis

Thesis (Charlot). Mainstream AI economics often treats demand for intelligence like demand for oil: elastic, unbounded, forever hungry. For **work and care**, that is the wrong physics. Humans **satiates**. When a loop is *done*—invoice filed, child picked up, patient assessed, pipeline green—extra synthesis does not create human value. It creates noise, cost, and overkill. Capability can scale. Appetite cannot.

Five findings (evidence-weighted).

1. **Satiation is a real economic object**, not a soft preference. Classical consumer theory already admits **bliss points** where more of a good reduces utility (see §1). Engagement KPIs (tokens, seats, time-on-task) systematically diverge from **economic done**.
2. **Tech cycles really do race** → **satiate** → **commodity**, but the analogies are strongest as **[SOFT]** pattern recognition, not law. Verifiable cases (1890s bicycle boom/bust; U.S. smartphone ownership flat at ~91%; LCD panels as commodity; airline seats as yield-managed commodities) support the narrative where labeled.
3. **Digital AI is already on a free-at-margin arc** for many chores. Epoch AI documents extreme inference price declines for fixed performance (~9–900x/yr in Mar 2025 insight; ~47%/quarter ≈ 13x/yr in Sep 2026 “plunging price of thought” report). Open weights and System One–style typed decisions further cannibalize synthesis scarcity (§4). This is **not** free joules—physics still meters energy.
4. **Care and work completeness** are where satiation is most obvious and most ignored. Baumol’s cost disease literature shows why labor-intensive care resists pure productivity theater; Autor / Acemoglu–Restrepo task frameworks show automation displaces *tasks*, not “infinite appetite.” More AI after done can harm (§5). Steelman counters (status, creative unbounded demand, military/R&D) are real and scoped (§5.4).
5. **What does not discount**: physics (Landauer lower bound; Horowitz CMOS energy reality), actuators/matter, liability, and **commit**. OpenIE meters joules; WCA refuse-to-commit makes economic done U physical unsafe both refuse (§6–7).

Deck line (unchanged from blueprint): Capability scales. Appetite does not. Design to done.

Product one-liner: Sell completion. Meter joules. Refuse after the finish line—and refuse before the plant moves without a certificate.

1. Definitions

1.1 Satiation

Economic satiation (standard): the more of a good one already has, the less one is willing to give up for more—driven by diminishing (and eventually zero / negative) marginal utility. Wikipedia’s summary and Andersen (2001) frame this as a structural feature of preference dynamics, not merely “taste.”

Bliss point (textbook consumer theory): a bundle $((x^*, y^*))$ at which further increases in consumption *reduce* utility. Indifference curves become closed around the bliss point; the usual monotonicity (“more is better”) fails. If the bliss point is affordable, the consumer stops there and may not spend all income. See Varian-adjacent textbook treatments (e.g. LibreTexts / *Introduction to Economic Analysis* §12.4 examples on pizza–beer bliss points).

Operational (this study / OpenIE–WCA): A task or care loop is **satiated** when additional synthesis, tooling, tokens, or “AI features” do not increase **human-valued completeness** (work closed / care closed). Past that threshold, more AI $\approx$ noise + cost + attention tax.

This operational definition is deliberately narrower than “all goods eventually satiate.” Status goods, military capability races, and open-ended R&D can remain non-satiating (§5.4). The claim is about **work & care completeness**, not about every human preference.

1.2 Completeness

Term	Meaning here
Work completeness	Ticket closed, patch shipped, report filed, ledger reconciled, acceptance criteria met.
Care completeness	Assessment finished, meds reconciled, family notified, child safe, elder transfer done.
Economic done	Human-valued finish line for that loop—orthogonal to model capability ceiling.
Technical capacity	What the system <i>can</i> still generate after done.

Completeness is a **stop condition**, not a quality score. A done detector that fires too early is under-care; one that never fires is engagement cosplay.

1.3 Economic done vs engagement KPIs

Lens	Engagement / unbounded model	Satiation / completeness model
Demand curve	Always more intelligence useful	Flat (or negative) after completeness
Product KPI	Engagement, tokens, seats, sessions	Time-to-done; refuse rate after done; joules after done
Pricing story	Scarcity premium forever	Commodity after satiation; premium where physics / commit bind
Failure mode	Under-build capacity	Overkill after closed loops
Human constraint	Assumed elastic attention	Finite attention, trust, hours, responsibility
Safety story	Soft confidence / more monitoring	Typed refuse + hold at commit boundary

Attention economy contrast. Tristan Harris / Center for Humane Technology critique engagement maximization as extracting attention rather than serving user goals (humanetech.com; Harris talks / 80,000 Hours interview). Tim Wu’s *The Attention Merchants* (Penguin / Knopf lineage) frames attention as inventory sold to advertisers. **Implication for AI product:** regenerating after done is the enterprise cousin of infinite scroll—same KPI pathology, different costume.

1.4 Contrast with unbounded-demand AI forecasts

Unbounded-demand models treat intelligence-as-a-service like oil: more capability $\Rightarrow$ more willingness-to-pay $\Rightarrow$ larger TAM forever. That can be approximately true for:

- new task categories (reinstatement in Acemoglu–Restrepo sense);
- status / arms-race domains;
- frontier R&D where “better” has no finish line.

It is a **category error** for satiated digital chores (spell-check-shaped synthesis, closed tickets, closed care assessments). Goldman Sachs Research’s *Gen AI: Too Much Spend, Too Little Benefit?* (Jun 2024; overview <https://www.goldmansachs.com/insights/top-of-mind/gen-ai-too-much-spend-too-little-benefit> ; PDF mirrors circulated as TOM_AI 2.0) is an existence proof that serious sell-side research already doubts whether ~\$1T-scale infra spend maps cleanly to economy-wide productivity.

Secondary coverage (e.g. 404 Media summary of the same note) amplifies the skepticism; treat magnitudes as **[VENDOR/SELL-SIDE]** and re-check the primary PDF before quoting numbers in a whitepaper.

Category error to avoid: “AI gets cheaper $\Rightarrow$ infinite demand.” Jevons / rebound say cheaper energy *services* can increase *use*—but only where appetite is not already at a bliss point. For closed chores, cheaper AI often means **same done, lower bill**—or **same done, more noise if loops don’t stop**.

2. Literature & adjacent fields

2.1 Satiation in consumer theory / utility

Source	Claim	URL / ID
Andersen, Esben Sloth (2001). “Satiation in an Evolutionary Model of Structural Dynamics.” <i>Journal of Evolutionary Economics</i> 11(1): 143–164	Evolutionary / structural dynamics with satiation	DOI: 10.1007/PL00003852
Wikipedia <i>Economic satiation</i>	Survey stub linking satiation to diminishing MU	https://en.wikipedia.org/wiki/Economic_satiation
LibreTexts / <i>Introduction to Economic Analysis</i> §12.4	Bliss-point isoquants; satiation as max utility then decline	https://socialsci.libretexts.org/ (search “bliss point”)
Econ-Viz <i>Satiation (Bliss Point)</i>	Quadratic utility ($U = -la(x - x^*)^2 - lb(y - y^*)^2$); closed indifference ellipses	https://econ-viz.org/models/satiation/

Takeaway for AI econ: “More is better” is an *axiom choice*, not a law of nature. Work/care products that assume monotonicity past completeness will misprice and overbuild.

2.2 Jevons paradox & rebound

Source	Claim	URL
Jevons, W. S. (1865). <i>The Coal Question</i>	Efficiency gains can <i>increase</i> coal use (“backfire”)	Historical classic; modern reprints vary
Gillingham, Rapson & Wagner (2015/16). “The Rebound Effect and Energy Efficiency Policy.” <i>REEP</i>	Micro rebound often ~20–40%; little support for universal backfire; macro harder	https://gwagner.com/wp-content/uploads/Gillingham-Rapson-Wagner-2015-Rebound-Effect.pdf
Stern (2020 survey CAMA)	Economy-wide rebound estimates mixed; some large	https://cama.crawford.anu.edu.au/sites/default/files/publication/cama_crawford_anu_edu_au/2020-07/70_2020_stern.pdf
TypeSafe “Jev” naming	Explicit nod to Jevons paradox: cheaper decisions → more decisions	Product landscape context; [VENDOR] branding

Bridge to satiation thesis: Rebound requires residual demand elasticity. At a bliss point / completeness threshold, the direct rebound on *that chore* collapses toward zero—you do not drive infinite miles once you have arrived; you do not spell-check forever once the document is correct. Rebound can still appear as **new tasks** (Acemoglu–Restrepo reinstatement) or **attention spillover** (agents inventing work). Product rule: meter post-done joules separately from pre-done joules.

2.3 Baumol cost disease / care economy

Source	Claim	URL
Baumol, W. J. (1967). “Macroeconomics of Unbalanced Growth.” <i>AER</i> 57: 415–426	Progressive vs stagnant sectors; relative cost rise in labor-intensive services	Classic AER cite
Hartwig / OECD panel literature	Baumol effect statistically linked to health spending growth	e.g. Hartwig <i>J Health Econ</i> 2008 lineage
Colombier & Hartwig / related (2025) acute vs LTC	Baumol affects both acute and long-term care; stronger in LTC	https://link.springer.com/article/10.1007/s10754-025-09392-9 ; PMC https://pmc.ncbi.nlm.nih.gov/articles/PMC12361285/
Bates & Santerre / SSQ	Cost disease ~15–40% of “full” effect on HCE in one instrumented design	

Source	Claim	URL
		https://onlinelibrary.wiley.com/doi/10.1111/ssqu.12384
Atella et al. / <i>Health Economics</i> (2018)	Skeptical: little support for BCD once trending addressed	https://onlinelibrary.wiley.com/doi/10.1002/hec.3641
Pomp & Vujić (CPB)	~0.5% real health spend growth per 1% economy-wide productivity	https://www.cpb.nl/system/files/cpbmedia/publicaties/download/rising-health-spending-new-medical-technology-and-baumol-effect.pdf

Honesty: Baumol is contested in magnitude. Consensus-enough for this study: **care remains labor-, trust-, and liability-intensive**; AI that speeds documentation without a done/refuse discipline can raise measured “activity” while missing completeness—or worse, add generative noise into clinical workflows.

2.4 Automation: displacement vs task completion

Source	Claim	URL
Acemoglu & Restrepo (2019). “Automation and New Tasks.” <i>JEP</i> 33(2): 3–30	Automation = displacement; new tasks = reinstatement; “so-so” automation worst	https://www.aeaweb.org/articles?id=10.1257/jep.33.2.3 ; NBER https://www.nber.org/papers/w25684
Acemoglu & Restrepo (2018). “The Race between Man and Machine.” <i>AER</i>	Factor shares / employment under automation vs new tasks	https://pubs.aeaweb.org/doi/pdf/10.1257/aer.20160696
Author task literature (broader)	Polarization / routine-biased change; tasks not jobs	Multiple papers; use Author’s survey pieces when citing in whitepaper

Mapping: Completeness-oriented AI is closer to **closing a task** than to inventing engagement. Unbounded agent loops without done detectors look like “so-so automation”: costly activity, weak productivity.

2.5 Attention economy critiques

Source	Claim	URL
Center for Humane Technology	Engagement metrics extract attention; redesign incentives	https://www.humanetech.com/youth/the-attention-economy

Source	Claim	URL
Tristan Harris (80,000 Hours / talks)	Replace time-spent with time-well-spent; incentive redesign	https://80000hours.org/podcast/episodes/tristan-harris-changing-incentives-social-media/
Tim Wu, <i>The Attention Merchants</i>	Attention as sold inventory	https://www.penguinrandomhouse.com/books/234876/the-attention-merchants-by-tim-wu/

AI parallel: Token meters and seat licenses optimized for usage will fight satiation. Joule meters and done detectors align with completeness.

2.6 “Good enough” technology / Christensen commodity

Source	Claim	URL
Christensen, <i>The Innovator’s Dilemma</i> (1997)	Overshoot → customers become overserved; “good enough” wins on convenience/price	https://en.wikipedia.org/wiki/The_Innovator%27s_Dilemma ; Christensen Institute https://www.christenseninstitute.org/theory/disruptive-innovation/
Christensen Institute checklist	Disruption ≠ breakthrough sustaining innovation	same

Mapping [SOFT]: Frontier AI is sustaining innovation for many digital chores; open weights / small specialists / typed decision heads are “good enough” disruptors. After satiation, customers do not want louder capability—they want cheaper, quieter completion.

2.7 Historical tech diffusion cycles (verifiable notes)

Cycle	Verifiable anchor	Honesty
Bicycles (1890s)	British Bicycle Mania 1895–1900: cycle shares +>200% then →75% (Quinn & Turner, <i>Cambridge Journal of Economics</i> 2019)	Strong on boom/bust finance; satiation-of-use is [SOFT] reading
	U.S. peak sales per capita ~1897 not matched until ~1965 (Dowell & Swaminathan lineage; industry structure lit)	Strong on market saturation of prestige phase
Cars	Mass mobility / used markets after horsepower theater	[SOFT] narrative; cite specific series in whitepaper if needed
Planes / airlines	Post-1978 U.S. deregulation → seats as perishable commodities; yield/revenue management	Strong on commodity endgame of <i>seats</i> ; not on “speed records satiation”

Cycle	Verifiable anchor	Honesty
TVs	Developed-market ownership near universal; LCD panels traded as commodities; replacement cycles	Strong on panel commodity; resolution arms race → “good enough” is [SOFT]
Phones	Pew: U.S. smartphone ownership 91% in 2024 and 91% in 2025; cellphones 98%	Strong saturation evidence (https://www.pewresearch.org/internet/fact-sheet/mobile/)

2.8 AI TAM / forecast critiques

Source	Claim	Honesty
Goldman Sachs <i>Gen AI: Too Much Spend, Too Little Benefit?</i> (2024)	Infra spend vs benefit skepticism	Sell-side; verify PDF numbers before whitepaper quotes
Epoch AI inference price work (2025–2026)	Extreme price/performance declines—consistent with commodity phase for <i>capability-at-price</i> , not with infinite WTP	Research nonprofit; strong on cost curves
System One wave (Jev / Laya, Sep 2026)	Typed decisions collapse generation tax; moats thin	See PRODUCT_LANDSCAPE.md; [VENDOR] multiples

3. Cycle table deepened (race → satiation → commodity)

Technology	Race phase	Satiation signal	Commodity endgame	Evidence notes	Honesty
Bicycles	Novelty, clubs, prestige machines; 1890s mania	“I can get there”; middle-class craze exhausts	Utility transport / recreation kit; exports of surplus	Quinn & Turner (2019) share boom/bust; U.S. ~1897 peak	Race/finance strong ; “completeness” reading [SOFT]
Cars	HP wars, brand theater	Reliable trips > endless upgrades	Mass mobility, used markets	Pattern recognition across auto history	[SOFT]
Planes	Heroic aviation,	Arrive safely on a schedule	Seats + fuel + RM math	Airline RM / deregulation literature	Seat commodity strong ;

Technology	Race phase	Satiation signal	Commodity endgame	Evidence notes	Honesty
	speed records				heroics→schedule [SOFT]
TVs	Screen size / resolution arms race	Enough fidelity for content	Commodity panel + subscription pipe	Panel pricing reports; near-universal ownership in rich markets	Panel commodity strong ; “enough fidelity” [SOFT]
Cell phones / smartphones	Spec sheets, camera Mpx	Always-reachable communication	Pocket utility computer	Pew 91% smartphone / 98% cellphone (2025)	Saturation strong
AI (digital chores)	Frontier seats, agents, AGI theater	Chore closed without theater	Bundled / free-at-margin synthesis	Epoch cost curves; open weights; System One	Cost collapse strong ; satiation of <i>chores</i> is thesis (operational)

Rule (blueprint, unchanged): If your AI wedge is still “more capability” after the user’s finish line, you are selling race-phase theater into a commodity market.

Wildcard (~6 months frontier distill onto inferior hardware) [SOFT + emerging evidence]:

Distillation, quantization, BitNet/TeLLMe/T-MAC-class methods, and open cascades move yesterday’s frontier competence onto weaker silicon (PRODUCT_LANDSCAPE.md Axis B). Treat “~6 months” as a **heuristic planning horizon**, not a measured law—label **[UNVERIFIED]** as a universal constant; cite specific model/hardware pairs when claiming in whitepaper.

4. Free-at-margin arc (real cites only)

4.1 Spell-check → feature → air

Stage	Pattern	Cite
Product	Standalone spelling tools / early WP features	Wikipedia <i>Spell checker</i> https://en.wikipedia.org/wiki/Spell_checking
Feature	WordPerfect mid-1980s integration; MS Word background check (Word 95 red underline narrative)	Same; secondary histories (e.g. Inkbot editing history piece—use as [SECONDARY])
Air	Expected correctness layer around the text box; not sold as a seat	Lived infrastructure; hard to cite a single paper—treat as [SOFT] industry observation

Decision rule: Do not build a seat license for air.

4.2 Search, maps, open internet

Search and maps followed advertising / bundling paths toward “just there” marginal access. Exact pricing histories are firm-specific; the **pattern** (scarce discovery → expected infrastructure) is **[SOFT]** but widely observed. Prefer primary company filings or academic IO papers if the whitepaper needs hard numbers.

4.3 Open weights & inference cost curves

Finding	Source	URL
Price to hit fixed LLM performance fell ~ 9x–900x per year across six benchmarks (GPT-4-level GPQA Diamond ~ 40x/yr)	Epoch AI data insight, Mar 12, 2025	https://epoch.ai/data-insights/llm-inference-price-trends
Cost of given performance fell ~ 47% per quarter (~13x/yr) since ~2023; faster near SOTA	Epoch AI, <i>The plunging price of thought</i> , Sep 22, 2026	https://epoch.ai/publications/the-plunging-price-of-thought
Open-weight token share rising; revenue share lagging (press summaries Aug/Sep 2026)	Tech press / gateway reports	Treat specific % as [PRESS] until primary dashboards cited
System One typed decisions: collapse generation tax for known option sets	PRODUCT_LANDSCAPE.md; Laya catalog	https://laya-ai.com/system-one-models ; Jev [VENDOR]

Interpretation for satiation thesis:

- Free **at the margin** = priced like spell-check for pure information chores with clear completeness—not zero energy bill.
- Moats that were only **synthesis scarcity** get eaten by rivals, open weights, cascades, retrieval, typed heads.
- Frontier still matters for non-satiating domains (R&D, arms races, novel tasks)—do not overclaim.

4.4 People frontload payment **[SOFT operational]**

Users and firms often **prepay** (subscriptions, reserved capacity, seat licenses) while expecting AI to behave like air. That mismatch—frontloaded spend against satiating appetite—creates waste visible only if you meter **joules after done**. OpenIE’s job is to make that waste legible.

5. Care & work completeness

5.1 Where more AI after done harms

Domain	Completeness test	Harm of post-done AI
Office / tickets	Closed with acceptance	Re-prompt churn, latency, attention tax, shadow IT agents inventing work
Coding chores	Verified correct artifact	Regenerating lawful answers via frontier (CFC chemistry)
Healthcare documentation	Note complete, orders reconciled	Hallucinated embellishment, alert fatigue, billing narrative inflation
Eldercare / childcare logistics	Person safe, handoff done	Engagement-style chat replacing presence; false motion
Clinical decision support	Assessment + plan signed	Probability theater without liability-bearing commit

Baumol lens: the scarce input is often **trusted human time and responsibility**, not tokens. AI that cannot stop competes with that scarce input.

5.2 Healthcare / eldercare / childcare — completeness first

Design principle: **care is not a funnel**. Metrics should be time-to-safe-state, handoff integrity, and refuse-on-complete—not session length.

Empirically, health spending drivers are multi-causal (Baumol + tech + income + demographics). This study does **not** claim AI “solves Baumol.” It claims AI without satiation discipline can **worsen** measured busy-ness in stagnant-sector work.

5.3 Office work — done detectors before louder models

Blueprint rule, reinforced: **Ship a done detector before shipping a louder model**.

Acemoglu–Restrepo: prefer automations with real productivity (not so-so), and expect new tasks—but new tasks are not an excuse for infinite loops on *old* closed tasks.

5.4 Steelman counterarguments (non-satiating domains)

Counter	Steelman	Reply (scoped)
Status goods	Veblen / positional demand never satiates; “best model” is a trophy	True for status. False for invoice filing. Segment markets.

Counter	Steelman	Reply (scoped)
Creative unbounded demand	Art, research ideation, entertainment want more novelty	Partially true. Still often has <i>project-level</i> done (ship the album, submit the paper).
Military / security	Adversary sets the ceiling; arms races lack bliss points	True. Commit/safety gates still bind; satiation thesis does not apply cleanly.
Open-ended R&D / AGI race	Better science tools raise ambition	True for frontier labs. Does not rescue TAM math for satiated SaaS chores.
Jevons on AI	Cheaper inference $\Rightarrow$ vastly more inference	Often true <i>globally</i> ; can coexist with <i>per-chore</i> satiation. Meter both.
Engagement is the product	Some businesses sell attention	Then they are not in the work/care completeness business—say so honestly.
Humans don't know done	Preferences are constructed; more features reveal demand	Sometimes. Still need stop conditions to avoid harm in care/liability domains.

6. Scarcity residual: physics, joules, actuators, liability, commit

6.1 What stays scarce vs what commodities

Still scarce	Heading commodity
Joules / thermodynamic & CMOS energy cost	Generic text synthesis for closed chores
Matter, actuators, clinic/floor time	Spec-sheet “AI features” without completion
Irreversible commit / liability	Soft moats that were only temporary synthesis scarcity
Auditable certificates + refuse taxonomy	Confidence strings as safety
Honest meters (OpenIE joules)	Seat licenses with no ledger

6.2 Landauer (honesty)

Landauer (1961), “Irreversibility and Heat Generation in the Computing Process,” *IBM Journal*: logically irreversible operations require a minimum heat generation on the order of (kT) per irreversible bit operation (often quoted as (kT) per erased bit). PDF mirror: https://cqi.inf.usi.ch/qic/61_Landauer.pdf

Honesty for OpenIE messaging:

- Landauer is a **lower bound**, many orders of magnitude below practical CMOS dissipation.
- It does **not** mean today's LLM inference is near the bound.
- It **does** mean “information is free” is false in physics—erasure and irreversible commits have a thermodynamic story.

- Do not slap Landauer on a marketing slide as if GPT inference were (kT) away from free.

6.3 Horowitz / CMOS energy reality

Mark Horowitz, “Computing’s Energy Problem (and What We Can Do About It),” ISSCC 2014, pp. 10–14 (IEEE: DOI 10.1109/ISSCC.2014.6757323). Core message in secondary teaching notes and talks: systems are increasingly **energy-limited**; architectural specialization and reducing memory/IO energy matter more than wishing Dennard scaling back.

Honesty: Quote ISSCC paper carefully; vendor “X tokens/J” claims are **[VENDOR]** until DUT-metered. WCA stack: board_synth_claimed=false; surrogate J = OpCounter × analytical (E_*) **([SURROGATE])**.

6.4 Actuators, liability, commit gates → OpenIE + WCA

Compose (software reference, unchanged):

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commit = LUT allow ∧ energy/safe ok [∧ CBF]
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Signal	Gate behavior
Economic done (satiated chore)	Policy / budget refuse — do not spend
Physical unsafe / over-budget	Energy / LUT / CBF refuse — hold plant
Irreversible tool	No executor call until Certificate allows

Refuse = economic done U physical unsafe — two faces of one discipline.

OpenIE: **Measure · route · eliminate excessive waste** — meter in joules (or honest Wh), not tokens/seats/vibes as primary scarcity.

Linkage to notational intelligence research: certificates as **runtime commit law**, not soft prose (NOTATIONAL_INTELLIGENCE_RESEARCH.md).

7. Decision rules / product implications (extends blueprint)

7.1 Core rules (from blueprint, retained)

1. Design to **done**; sell completion.
2. Assume soft synthesis moats get eaten.
3. Plan for competence sliding down the device stack.
4. Compete on **meter + commit**, not chat theater.
5. Never claim “free” for energy or plant motion.
6. Ship **done detector** before louder model.
7. Probability routes; **certificates commit**.

7.2 Extensions from this study

ID	Rule	Rationale
R-S6	Split meters: pre-done J vs post-done J	Makes satiation visible; ties to rebound vs bliss-point distinction
R-S7	Segment TAM: satiating chores vs non-satiating races	Avoids applying AGI blue-ocean prices to spell-check-shaped work
R-S8	Treat engagement KPIs as hostile in care/work products unless proven aligned	Attention-economy literature
R-S9	For care: optimize handoff integrity / safe-state time, not session length	Baumol + completeness
R-S10	When citing Landauer/ Horowitz, state measurement tier (A surrogate / B post-synth / C board)	Honesty; matches NI research memo
R-S11	Jevons naming (cheaper → more) is a warning , not a growth excuse for uncapped agents	TypeSafe irony + rebound lit
R-S12	Kill criteria K-S1–K-S5 remain binding (blueprint §9)	Strategy hygiene

7.3 Anti-patterns (extended)

Anti-pattern	Why it fails
AGI blue-ocean price on satiated chores	Appetite already flat
Soft confidence as plant permission	Probability ≠ certificate
Token/\$ as only scarcity meter	Misses joules and irreversible commit
Infinite agent loops as “productivity”	No completeness stop
“Free AI” erasing energy/plant cost	Physics does not discount
Using rebound/Jevons to deny satiation	Rebound needs residual elasticity
Landauer cosplay (“we’re near kT”)	Orders of magnitude gap; [SURROGATE] risk

8. Implications for whitepaper + Medium #2 revision notes

8.1 What Medium #2 already covers

- Unbounded-demand critique; satiation definition for work/care
- Soft cycle table (bikes→cars→planes→TVs→phones)
- Free-AI arc (spell-check → internet → free at margin)
- Care & work completeness
- Physics + commit scarcity; OpenIE meters + WCA refuse
- CFC chemistry metaphor; compose formula

8.2 What this study adds (do not dump wholesale into Medium)

1. **Formal definitions** linked to bliss-point / consumer theory and engagement-KPI contrast.
2. **Literature map** with real DOIs/URLs (satiation, rebound, Baumol, Acemoglu–Restrepo, Christensen, attention economy).
3. **Evidence-labeled cycle table** (Quinn & Turner; Pew; airline RM; honesty labels).
4. **Quantitative free-at-margin spine** (Epoch 2025 insight + 2026 plunging-price report).
5. **Steelman counter-table** for status / creative / military / R&D / Jevons.
6. **Landauer + Horowitz honesty** (bound vs CMOS practice; measurement tiers).
7. **Extended decision rules** R-S6–R-S12 and whitepaper outline hooks.
8. **Bibliography + method appendix.**

8.3 Suggested Medium #2 micro-edits (optional; do not rewrite)

1. After the cycle table sentence “Soft reading of history, deliberately,” consider adding: “Deeper sources and honesty labels live in the companion research study.”
2. Where “Landauer’s ledger” appears, a footnote-level honesty clause helps: lower bound, not a claim that LLMs are near (kT).
3. Optional one-liner citing Epoch on inference price collapse (with link)—only if Medium style allows outbound research links.
4. No change needed to the compose formula or OpenIE/WCA positioning; study confirms, does not contradict.

8.4 Whitepaper skeleton (future)

1. Problem: AI econ without satiation
 2. Theory: bliss points, completeness, rebound scoped
 3. Empirics: cost curves + diffusion cases
 4. Care/work sector analysis
 5. Scarcity residual + commit nomenclature
 6. OpenIE + WCA reference architecture
 7. Falsifiers / kill criteria
 8. Full bibliography
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9. Bibliography (real URLs / DOIs only)

Economics & theory

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Internal companions

1. artifacts/presentations/MEDIUM_SATIATION.md

2. artifacts/SATIATION_BLUEPRINT.md

3. artifacts/PRODUCT_LANDSCAPE.md

4. artifacts/NOTATIONAL_INTELLIGENCE_RESEARCH.md

10. Honesty appendix + method

10.1 Method

Item	Detail
Dates	

Item	Detail
	Research & drafting Tue Sep 29–Wed Sep 30, 2026, America/ New_York (EDT, UTC−4). Box date at drafting: Tue Sep 29 23:11 EDT 2026.
Tools	WebSearch, WebFetch on primary pages; read of local Medium #2, blueprint, PRODUCT_LANDSCAPE, NOTATIONAL_INTELLIGENCE_RESEARCH.
Prior artifacts	Extended, not contradicted; contradictions would be flagged—none material found.
Citation rule	Real URL/DOI or tag [UNVERIFIED]/[VENDOR]/[SURROGATE]/[SOFT]/[PRESS]/[SECONDARY] .
Excluded	Fabricated papers, invented statistics, silicon joule leadership claims, board synthesis claims.

10.2 Label legend

Tag	Meaning
[SOFT]	Historical / narrative analogy; pattern-helpful, not econometric law
[UNVERIFIED]	Plausible but not pinned to a primary measurement here
[VENDOR]	Company self-report or branded benchmark
[SURROGATE]	Analytical / OpCounter energy, not DUT watts
[PRESS]	News secondary; re-check primary before whitepaper
[SECONDARY]	Blog/textbook exposition, not original paper

10.3 Explicit non-claims

- Does **not** claim all human demand satiates.
- Does **not** claim Landauer binds practical LLM energy.
- Does **not** claim WCA board joules (board_synth_claimed=false).
- Does **not** claim Baumol is the sole driver of health costs.
- Does **not** claim Epoch rates extrapolate forever.
- Does **not** treat Goldman Sachs as gospel—sell-side critique only.

10.4 Falsifiers (research-facing)

Falsifier	Would weaken thesis if...
F1	Controlled studies show post-completeness generative AI raises validated care/work outcomes without added harm
F2	Persistent WTP premiums for frontier synthesis on clearly closed chores despite open-weight parity

Falsifier	Would weaken thesis if...
F3	Inference prices stop falling <i>and</i> demand for closed-chore synthesis remains highly elastic for years
F4	Commit/liability becomes cheaply insurable at scale with soft scores alone (no certificates)

*End of study. Companion publish prose: Medium #2. Decision tables: SATIATION_BLUEPRINT.md.
OpenIE · WCA lane.*